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Pacific Blue-eye *Pseudomugil signifer*, Mossman Gorge, Qld. Photo: Jason Sulda

Blue-eye special issue

The Mangrove Blue-eye, a new species of *Pseudomugil* (Pisces: Pseudomugilidae) from eastern Queensland, Australia

Michael P. Hammer[1], Gerald R. Allen[2], Mark Adams[3,4] & Peter J. Unmack[5]

[1] Natural Sciences, Museum and Art Gallery of the Northern Territory, GPO Box 4646, Darwin NT 0801, Australia, michael.hammer@magnt.net.au

[2] Department of Aquatic Zoology, Western Australian Museum, Locked Bag 49, Welshpool DC, Perth, WA 6986, gerry.tropicalreef@gmail.com

[3] Evolutionary Biology Unit, South Australian Museum, North Terrace, Adelaide, SA 5000, Australia

[4] School of Biological Sciences, University of Adelaide SA 5005 Australia

[5] Centre for Applied Water Science, Institute for Applied Ecology, University of Canberra ACT 2617, Australia

http://zoobank.org/urn:lsid:zoobank.org:pub:B7620E97-EF44-4431-ACB2-B6C9ECE6AB09

Abstract

Pseudomugil halophilus n. sp. is described from 340 specimens, 9.0–32.3 mm SL, collected from mangrove habitats of coastal eastern Queensland. Based on general appearance the new species is most similar to *P. signifer*, which ranges widely in coastal streams of eastern Australia and co-occurs with the new taxon, between the vicinity of Gladstone and Townsville, Queensland. However, the two species clearly differ on the basis of nuclear (allozymes) and mitochondrial (ATPase6 & 8) genetic data, shape of the male second dorsal and anal fins (rounded in *P. halophilus* and pointed in *P. signifer*), and male fin colouration. Namely, the second dorsal fin of *P. halophilus* possess considerable blackish pigment on the central portion of the fin compared to *P. signifer*, which has an intense black spot on the basal part of the anterior-most rays and blackish posterior margin, and the first dorsal fin of *P. halophilus* and middle portion of the anal fin are also darkly pigmented compared to the fins of *P. signifer*, which are mainly pale except for a dark posterior margin on the anal fin. The new species represents an important addition to knowledge of mangrove ecosystems, with further understanding of ecology and local conservation required.

Introduction

The family Pseudomugilidae or Blue-eyes is a diverse group of small fishes endemic to Australia and New Guinea, with most species found on either side of the Sahul Shelf, a submerged, but shallow region between the two land masses. Eight species have previously been recorded in Australia including: the freshwater wetland generalists *Pseudomugil gertrudae* Weber 1911 and *Pseudomugil tenellus* Taylor 1964; other narrow range endemics in specialised wetland habitat, namely *Pseudomugil mellis* Allen & Ivantsoff 1982 and *Scaturiginichthys vermeilipinnis* Ivantsoff, Unmack, Saeed & Crowley 1991, of coastal wallum heath and artesian desert springs, respectively; the euryhaline *Pseudomugil signifer* Kner 1866 stretching along coastal catchments of most of the east coast (plus Torres Strait and Weipa); and the estuarine to marine *Pseudomugil cyanodorsalis* Allen & Sarti 1983 and *Pseudomugil inconspicuus* Roberts 1978 (see Allen 1995; Allen et al. 2002). A recent new addition to the known Australian fauna is the swamp dwelling *Pseudomugil paludicola* Allen & Moore 1981 (see Lawler et al. this issue).

The presence of a Blue-eye possibly new to science was brought to our attention via two simultaneous pathways as part of broader ongoing studies into the systematics of the family. In 2016 a citizen scientist, Matt Lachlan, observed a *Pseudomugil* with unusual appearance and behaviour from mangrove habitats in Bowen, north Queensland and discussed this with Jeff Johnson (Queensland Museum). Jeff facilitated contact and live specimens were sent to the Museum and Art Gallery of the Northern Territory, Darwin for photography and genetic assessment. These fish did indeed appear distinctive visually, being most similar to *P. signifer* but with short, rounded fins and distinctive fin colour patterns (horizontal black band of colour through both dorsal fins compared to more vertical pattern along fin edges with the fins extended in *P. signifer*). Secondly, around the same time (mid-2016), genetic screening at the South Australian Museum (allozymes) of research samples from the Gladstone region via Ben Cook collected in 2007 and from two sampling trips undertaken by Peter Unmack targeting *P. signifer* across central coastal Queensland in 2014 and 2015, identified anomalous *P. signifer* individuals that were clearly distinct but still a Blue-eye ingroup (i.e. likely new species). This result was confirmed by further allozyme screening and later independently confirmed using mitochondrial DNA. Retrospective examination of preserved companion voucher material from the 2014/2015 field collections, and other museum material, was then undertaken to assess whether the 'mangrove' species was new to science by using a combined lines of evidence approach (genetics, morphology and biology).

With respect to the appropriate comparisons for the new species, elevation of an older synonym of *P. signifer* has been touted based on the presence of two divergent mitochondrial lineages north and south of the Burdekin River; *Pseudomugil signatus* Günther 1867 applying to the northern lineage (McGlashan & Hughes 2002; Wong et al. 2004). However, the mitochondrial data appears at odds with nuclear genetics and morphology which suggests a single variable taxon (Hadfield et al. 1979; Saeed et al. 1989). Further assessment of the diagnosability of these lineages is beyond the scope of the current study; however, we include material of both mitochondrial lineages within the zone of overlap with the new species allowing a brief review of the issue.

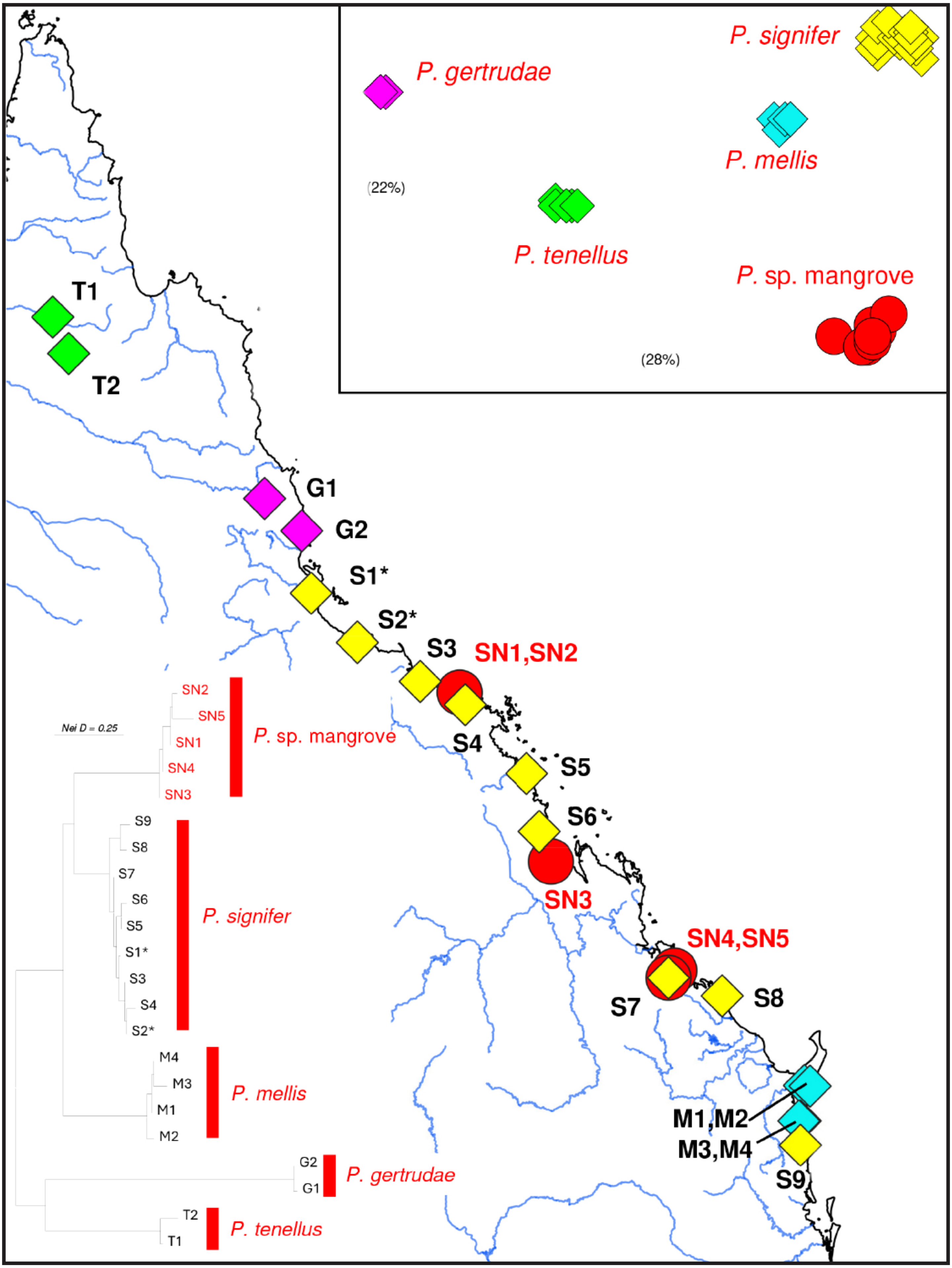

Figure 1. Map depicting the geographic location of all sites surveyed in the allozyme analyses. Site codes match those used in Table 1. The top inset displays the Principal Co-ordinates Analysis on all individuals, with the percentage contribution of the first two dimensions shown in backets. The bottom inset displays the NJ tree among sites, rooted at the mid-point.

Table 1. Taxon, site, and sampling details for all *Pseudomugil* examined in the allozyme study. All localities are in Queensland (N = sample size for each site). Sites prefix coded according to species (SN = *P*. sp. mangrove, S = *P. signifier*, G = *P. gertrudae*, M = P. mellis, and T = *P. tenellus*). *Pseudomugil signifer* sites north of the Burdekin River are marked with an asterisk. All samples are from the Australian Biological Tissue Collection (ABTC) at the South Australian Museum.

Site	ABTC code	Locality	Latitude	Longitude	N
SN1	PU14-102	upper Sand Hills Creek, Bowen	-19.9915	148.2479	4
SN2	MH-Bowe	mangrove creek, Bowen	-19.993	148.247	3
SN3	PU14-91	St Lawrence River	-22.3409	149.5254	2
SN4	PU15-17	Auckland Creek, Gladstone	-23.8571	151.2436	2
SN5/S7	BC07-m02	Calliope River	-23.958	151.159	1/2
G1	GB06-m03	Leslie Creek, Yungaburra	-17.298	145.534	3
G2	GB06-m04	Stagnant Creek, El Arish	-17.748	146.051	3
M1	MHXX-m01	aquarium stock, ex Fraser Island	-25.45	153.05	2
M2	GU94-m01	Lake Wabby, Fraser Island	-25.458	153.13	2
M3	LOR09-m02	Teewah Creek, Tin Can Bay	-25.943	152.993	1
M4	LOR09-m03	Snapper Creek, Tin Can Bay	-25.944	152.9745	3
S1*	PU15-25	Herbert River, Gairloch	-18.617	146.1831	2
S2*	PU14-106	Ross River tributary, Townsville	-19.2956	146.8246	4
S3	PU14-103	Molongle Creek	-19.8385	147.6989	2
S4	PU14-101	Yeates Creek	-20.1624	148.3255	2
S5	PU14-115	Goose Pond outlet, Mackay	-21.1204	149.1831	2
S6	PU14-96	Tributary to Carmila Creek	-21.9272	149.3616	2
S8	PU15-14	Creek behind Agnes Water town beach	-24.2091	151.9069	2
S9	PU09-105	Cootharaba Lake	-26.2826	152.9959	2
T1	MH15-36	Edward River, floodplain channel	-14.7787	142.5793	3
T2	MH15-45	Crosbie Creek, floodplain wetlands	-15.2862	142.8008	3
					52

Materials and Methods

Allozyme analyses

We chose allozyme profiling to provide an independent multi-locus assessment of the taxonomic distinctiveness of *P*. sp. mangrove. Frozen tissues of 12 *P.* sp. mangrove from five sites were compared to exemplars representing the other four species of freshwater *Pseudomugil* found in eastern Australia i.e., *P. gertrudae*, *P. mellis*, *P. signifer*, *and P. tenellus*. All tissues were collected under valid Queensland collecting and ethics permits and were sourced from the SA Museum's Australian Biological Tissues Collection. No frozen tissues were available for *S. vermeilipinnis* at the time the study was undertaken.

The *P. signifer* sites were selected to enclose the geographic range of *P.* sp. mangrove. As a consequence, they also spanned the presumed geographic divide between the northern and southern lineages of *P. signifer*. Site and sampling details for all 52 tissues screened are presented in Table 1 and these sites mapped in Figure 1.

Allozyme electrophoresis of muscle homogenates was undertaken on cellulose acetate gels. Laboratory protocols, enzyme details, and the methods employed for carrying out Principal Co-ordinates Analysis (PCoA), calculating genetic distances/fixed differences, and generating a Neighbor-Joining (NJ) tree have been published previously (Richardson et al. 1986; Hammer et al. 2007; Adams et al. 2014).

Mitochondrial DNA analyses

We included a total of 40 individuals from the six Pseudomugilidae species (Table 2). We sequenced two mitochondrial genes (ATPase6 & 8) using the primers Lys.22F (AAAGCGTTAGCCTTTTAAGC) and CO3.23R (GGCTTGGGTCAACTATGTGGT) following the methods given in Allen & Unmack (2012). Sequences were analysed with maximum likelihood using IQ-TREE 2.2.2.7 (Minh et al. 2020) run on the CIPRES server (Miller et al. 2010) incorporating IQ-TREE's model selection procedure (-m TEST; Kalyaanamoorthy et al. 2017), resulting in the selection of the TVM+F model followed by 10,000 replicates of ultrafast bootstrapping with the SH-like approximate likelihood-ratio test (-aLRT 1000; Hoang et al. 2018). Trees were rooted with *P. gertrudae*. Among species variation was calculated using mean between group p-distances in MEGA 7 (Kumar et al. 2016). All sequences obtained in this study were deposited in GenBank, accession numbers PQ128274–PQ128309.

Morphological approach

Morphometric and meristic data for examining *P.* sp. mangrove were based on 26 specimens, 22.1–32.3 mm SL. More than 700 lots containing over 10,000 specimens of *P. signifer* were screened while searching for additional examples of the new taxon. These included material from the Australian Museum (466), Museum and Art Gallery of the Northern Territory (32), Queensland Museum (196) and Western Australian Museum (49). Comparative data for *P. signifer* is summarised from Saeed et al. (1989) who examined in detail populations from across the range in their previous revision of the genus.

Counts of fin spines are given in Roman numerals and soft (segmented) rays in Arabic. The last ray of the second dorsal and anal fins is divided near the base and counted as a single element. Gill raker counts refer to the total count on the first branchial arch, including rudiments. Other counts and measurements were taken as follows: standard length (SL) – measured from the tip of the upper lip to the caudal-fin base (hypural junction); head length (HL) – measured from the tip of the upper lip to the upper rear edge of the gill opening; body depth – greatest depth; body width – measured just posterior to the opercular flap; snout length – measured between the front of the upper lip and the anterior edge of the bony orbit; eye diameter – horizontal measurement across middle of eye; interorbital width – the least bony width; caudal peduncle depth – the least depth; caudal peduncle length – measured between two vertical lines, one passing through the base of the last anal ray and the other through the caudal-fin base; pre-first-dorsal, pre-second-dorsal, preanal, and prepelvic distances – measured from tip of upper lip to respective origins of first dorsal, second dorsal, anal, and pelvic fins; second dorsal and anal fin base lengths – measured between the base of the first and last fin-ray elements of each fin; pectoral and pelvic fin lengths – measured from the base to the tip of the longest ray; caudal fin length – measured from the base of the fin (hypural junction) to the tip of the longest ray; lateral scales – number of scales in horizontal row from scale immediately above pectoral-fin base to caudal-fin base (hypural junction); transverse scales – number of scales in vertical row between anal fin origin and base of dorsal fin; predorsal scales – number of scales along midline of nape in front of first dorsal fin; cheek scales – total number of scales covering the preoperculum below the eye; opercular scales – total number of scales covering combined opercle and subopercle (usually 2 large scales on opercle and 2–3 smaller scales on lower opercle and subopercle).

Results

Allozyme analyses

The final dataset comprised the allozyme profiles of 52 fish at 57 putative allozyme loci. These data are summarized by species in Table 3. The Principal Co-ordinates Analysis on all individuals and the NJ tree for sites (Fig. 1) both unequivocally resolved all five *Pseudomugil* species, including *P.* sp. mangrove, as distinct and well-separated clusters. All species were readily diagnosable from each other by at least 19 fixed allozyme differences (range 19–44), levels of distinctiveness that go well beyond yardsticks that typically diagnose congeneric species in allopatry (Richardson et al. 1986; Adams et al. 2014) and become definitive for co-occurring taxa (*P.* sp. mangrove and *P. signifer* at the Calliope River site; Table 1).

While limited in geographic scope, the allozyme data for *P. signifer* did not support the presence of a simple north versus south dichotomy in this species centred around the Burdekin catchment (herein sites S1 and S2 versus the rest; Fig. 1). Instead, the pattern in the NJ tree was more consistent with a north to south cline in genetic relatedness in this portion of the species' range.

Mitochondrial DNA analyses

The mitochondrial DNA dataset contained 848 base pairs per individual, with a total of 40 fish included from six species. Sequence analysis of the 40 individuals yielded 460 invariant characters, seven variable but parsimony uninformative characters, and 381 parsimony informative characters. Maximum likelihood analysis recovered one tree with a likelihood score of -4586.639 (Fig. 9). All species examined were clearly separated, with *P.* sp. mangrove differentiated from the other species by large p-distances of 21.5–27.2%. There was generally strong bootstrap support except the node leading to *P.* sp. mangrove; given the short length of DNA used it is not surprising that a clear sister group relationship for *P.* sp. mangrove is not strongly supported. Similar to the allozyme results (Fig. 1), it is placed as the sister species to *P. signifer*, but this should only be considered provisional until more robust data are available to further address this question.

Table 2. Species, museum code, locality, and sample size details for all species examined in the mitochondrial DNA study. All museum codes relate to the Australian Biological Tissue Collection at the South Australian Museum except the two Queensland Museum (QM) samples.

Species	Museum code	Locality	Latitude	Longitude	N
P. sp. mangrove	PU14-105	Small tributary to Haughton River	-19.4769	147.1019	2
	PU14-102	upper Sand Hills Creek, Bowen	-19.9915	148.2479	2
	MH-Bowe	mangrove creek, Bowen	-19.993	148.247	2
	PU14-91	St Lawrence River	-22.3409	149.5254	1
	PU15-17	Auckland Creek, Gladstone	-23.8571	151.2436	1
P. signifer	QM:38283	Mabuiag Island, Torres Strait	-9.95	142.18	1
	QM:38285	Moa Island, Torres Strait	-10.18	142.28	1
	GU07-m03	Anichs Bridge, Mossman River	-16.4503	145.3599	3
		Behana Creek, Mulgrave River	-17.1340	145.8327	1
		Ross River	-19.3011	146.7924	1
	BCXX-m01	Broken River, Burdekin River	-21.1678	148.5051	2
	LOR08-m01	Baffle Ck, nr Miriamvale	-24.3555	151.6128	3
P. mellis	MHXX-m01	aquarium stock, ex Fraser Island	-25.45	153.05	2
	GU94-m01	Lake Wabby, Fraser Island	-25.458	153.13	2
P. tenellus	PU01-13	Mary River billabong	-12.9048	131.6373	5
S. vermeilipinnis		Edgbaston, spring NE60	-22.7316	145.4146	1
		Edgbaston, spring NW30	-22.7316	145.4146	1
		Edgbaston, spring NW90N	-22.7316	145.4146	1
		Edgbaston, spring SW70	-22.7316	145.4146	1
P. gertrudae	GB06-m03	Leslie Creek, Yungaburra	-17.298	145.534	3
	GB06-m04	Stagnant Creek, El Arish	-17.748	146.051	3

Table 3. Allozyme profiles and pairwise genetic distance measures for sp. mangrove and the other four species screened. Invariant loci: Ak, Ald1, Gapd1, Gp1, and Ldh1. Sample sizes in brackets. A dash (-) indicates this locus was not interpretable in this species due to partial overlap with another expressed locus. Genetic distance matrix is placed at the end of the table: lower triangle = number of pairwise fixed differences; upper triangle = unbiased Nei's Distance.

Locus	***P.* sp. mangrove (12)**	***P. gertrudae* (6)**	***P. mellis* (8)**	***P. signifier* (20)**	***P. tenellus* (6)**
Acon1	-	b	f	e^{91},a^{9}	c^{75},d^{25}
Acon2	d	a^{75},b^{25}	f^{87},e^{13}	f^{97},c^{3}	c
Acon3	a^{92},i^{8}	g^{83},h^{17}	d^{75},e^{25}	d^{52},c^{45},b^{3}	f
Acp1	c	a	c	c^{97},b^{3}	d
Acp2	d	b	d^{94},e^{6}	d^{92},c^{5},a^{3}	b
Acp3	-	a	b	-	c
Acyc	b	c	b^{75},a^{25}	b^{92},a^{8}	d
Ada	f^{96},h^{4}	j	a	$i^{53},g^{22},j^{12},f^{5},b^{3},e^{3},c^{2}$	d
Adh	b	f^{75},d^{25}	c	b^{72},a^{28}	e^{50},f^{50}
Ald2	b	a	b	b	b
Ca1	a	c	b	b	d
Ca2	d	a	d	$d^{77},c^{18},e^{3},f^{2}$	b
Ck	a	b	a	a	b
Enol	b	c	b	b	a
Est	b	c	a	b^{77},a^{23}	b^{75},a^{25}
Fdp	b	a	b^{81},d^{13},a^{6}	b^{98},d^{2}	c
Fum	a	c	a	a	b
Gapd2	a	a	a	a	b
Glo	c	a	b	c^{97},b^{3}	c
Got1	c^{96},f^{4}	d^{92},e^{8}	c	a^{98},b^{2}	b
Got2	c	a	b	d	c
Gp2	a	b	b	a	a
Gp3	a^{92},c^{8}	b	c	c^{60},a^{40}	c
Gpd	a	d	a	a^{98},b^{2}	c
Gpi1	b^{96},d^{4}	e	b^{87},a^{13}	$d^{70},b^{15},e^{10},a^{5}$	c
Gpi2	d^{96},f^{4}	b	c	e^{90},c^{8},g^{2}	a
Gsr	c^{96},d^{4}	a	c	d^{80},b^{20}	c
Idh1	c^{86},b^{14}	c	c	d	c^{75},a^{25}
Idh2	d	c	a	$d^{75},b^{10},a^{8},e^{7}$	f
Lap	c	c	b	c	a
Ldh2	a^{83},b^{17}	c^{58},d^{42}	a	a	a
Mdh1	b	a	b^{88},d^{12}	b^{98},c^{2}	b
Mdh2	c	d^{92},b^{8}	a^{94},d^{6}	b^{55},d^{45}	d
Mdh3	a	b^{92},d^{8}	a	a	b^{50},a^{42},c^{8}
Me1	a	b	b	b^{85},a^{15}	b
Me2	b	d	b	b	c^{58},a^{42}
Mpi	e	f	b	d^{95},a^{5}	b^{50},c^{50}
Ndpk1	a^{96},b^{4}	c	c	c	c
Ndpk2	c	b	a	b	a
Np	b	b	b^{81},a^{19}	b	b
PepA1	d^{95},e^{5}	g	b^{63},c^{31},a^{6}	d^{98},e^{2}	f
PepA2	d	b	b^{75},d^{19},a^{6}	d^{85},c^{8},e^{7}	d^{92},f^{8}
PepB	h^{79},j^{13},k^{8}	k^{50},l^{50}	g^{88},i^{12}	$b^{75},d^{15},a^{8},c^{2}$	f^{58},e^{42}
PepD1	b	a	-	b	c
PepD2	e^{67},c^{33}	c^{58},d^{42}	g	c^{95},b^{3},e^{2}	f^{92},a^{8}
6Pgd	e^{96},h^{4}	$b^{42},c^{33},a^{17},f^{8}$	d	g^{97},e^{3}	g
Pgk	b	a	b	a^{63},b^{37}	b
Pgm	a	b	c	d^{87},c^{13}	d
Pk	c	a	b	b	c
Tpi1	b	b	a	a	b
Tpi2	b	c	b	b	b^{83},a^{17}
Ugpp	c	a	b	$e^{40},b^{33},f^{22},a^{5}$	d
P. sp. mangrove	-	1.64	0.68	0.48	0.98
P. gertrudae	44	-	1.50	1.36	1.36
P. mellis	26	44	-	0.52	1.08
P. signifier	19	41	19	-	0.97
P. tenellus	34	42	35	34	-

Table 4. Summary of morphological comparison between *Pseudomugil* sp. mangrove and *Pseudomugil signifer*. Data for *Pseudomugil mellis* is also included to highlight the general conserved morphology in the group (D2 = second dorsal fin; *meristic and morphometric data for summarised from Saeed et al. 1989; appearance information based largely on this study and Allen & Ivantsoff 1982, and is accessible primarily for males)

Trait	*P.* sp. mangrove	*P. signifer**	*P. mellis**
Sample size	26	43	19
Meristic			
First dorsal fin spines	4–7 (5.1)	3–6 (4.9)	4–5 (4.3)
Second dorsal fin segmented rays	7–9 (7.8)	6–10 (7.4)	5–8 (6.4)
Anal fin rays	9–12 (11.2)	9–12 (10.4)	8–11 (9.6)
Pectoral fin rays	10–12 (11.2)	8–13 (10.1)	8–11 (9.0)
Lateral scales	26–28 (27.2)	25–31 (27.8)	26–29 (27.2)
Transverse scales	5–6 (6)	5–6 (5.5)	5–6 (5.2)
Predorsal scales	10–13 (11.7)	9–12 (10.3)	8–12 (9.8)
Preopercle scales	2–3 (2.1)		3-4
opercular scales	4-6 (4.6)		3-4
Gill rakers total	11–14 (12.2)	9–12 (10.9)	10–13 (11.7)
Morphometric			
Head length (in SL)	3.4–4.1 (3.7)	3.3–4.1 (3.7)	3.5–4.4 (3.9)
Body depth (in SL)	3.5–4.6 (3.9)	3.5–4.4 (4.0)	3.4–5.2 (4.5)
Caudal peduncle depth (in SL)	7.7–9.7 (8.5)	8.4–11.6 (9·5)	8.8–11.1 (9.8)
Pre-first dorsal distance (in SL)	1.8–2.1 (1.9)	1·8–2.2 (2.0)	1.9–2.1 (2.0)
Eye diameter (in HL)	2.4–3.0 (2.7)	2.2–3.2 (2.8)	1.8–3.0 (2.6)
Interorbital width (in HL)	2.0–2.4 (2.3)	1.9–2.5 (2.4)	1.8–2.8 (2.5)
Caudal peduncle length (in HL)	1.1–1.4 (1.2)	0.9–1.6 (1.1)	0.1–1.1 (1.0)
Snout length (in ED)	1.1–1.6 (1.3)	1.1–2.2 (1.5)	1.2–1.9 (1.7)
*Appearance**			
Maximum SL (approximate)	33 mm	55 mm	30 mm
Shape of D2 and anal fins	rounded	pointed	rounded
Longest ray(s) of second dorsal fin	3rd–5th	2nd	2nd–3rd
Longest ray(s) of anal fin	4th–6th	2nd	2nd
Triangular black mark at front of D2	absent	present	absent
Inter-radial membranes of D2	dark	pale	pale
Margin of D2 and anal fin	pale (yellow)	dark	pale (white)

Morphological comparisons

A summary of key morphological traits of *P.* sp. mangrove compared to *P. signifer* is shown in Table 4. These data highlight an almost complete overlap in both meristic and morphometric ranges, with similar means for all characters. Assessment of visual appearance does confirm a qualitative morphological difference, namely the second dorsal fin rounded (with 3rd to 5th ray longest and lacking filaments) compared to pointed (with 2nd ray longest, and males often with long filamentous extensions). The noted colour difference also holds, most pronounced in males, observed as dark versus pale inter-radial membranes and with the margin of the second dorsal fin pale versus dark (including thickened triangular mark at anterior base) (compare Figs 3–7). *Pseudomugil signifer* also seems to grow much larger. The relatively subtle differences between *P.* sp. mangrove and *P. signifer* are similar to those observed between the *P. signifer* and *P. mellis* species pair (Table 4).

While both nuclear and mitochondrial genetic markers were unequivocal in distinguishing *P.* sp. mangrove as a distinct biological species, the more traditional morphological characters used in assessing species boundaries were conserved. Nevertheless, differences are observed in male colouration and fin shape that provide combined lines of evidence support for species description (below). Data in parentheses in the species description refer to range of values for paratypes followed by the mean value for all specimens. Type specimens are deposited at the Australian Museum, Sydney (AMS), Museum and Art Gallery of the Northern Territory, Darwin (NTM), Queensland Museum, Brisbane (QM), and the Western Australian Museum, Perth (WAM).

Pseudomugil halophilus n. sp.

Mangrove Blue-eye

(Figs 3–5,7; Tables 5–6)

Holotype: QM I.41389, male, 27.5 mm SL, Auckland Creek, Gladstone, Queensland, 23° 51.424′S, 151° 14.616′E, P. Unmack, R. Wallace & R. Remington, 19 May 2015 (PU15-17).

Paratypes: AMS I.34358-001, 77 specimens 19.0–31.5 mm SL, Georges Creek, Queensland, 22° 37.91′S, 150° 33.31′E, D. Hoese & party, 28 September 1993; AMS IA.6639, 2 specimens, 19.2–20.5 mm SL, Bowen, Queensland, 20° 01′S, 148° 15′E, G.P. Whitley, 16 August 1935; NTM S.17868-001, 2 specimens, 25.7–30.0 mm SL, Mackay Airport drain, Mackay, Queensland, 21° 09.79′S, 149° 11.21′E, P. Unmack, 13 September 2014 (PU14-99); NTM S.17871-001, 16 specimens, 10.6–23.1 mm SL, Sand Hills Creek, Bowen, Queensland, 19° 59.49′S, 148° 14.87′E, P. Unmack, 14

Table 5. Proportional measurements of *Pseudomugil halophilus* type specimens expressed in percentage of SL.

		Males				Females			
Character	**Holotype**	**Min.**	**Max.**	**Avg.**	**n**	**Min.**	**Max.**	**Avg.**	**n**
Standard length	27.5	23.4	30.1	27.4	15	22.1	32.3	26.3	11
Body depth	27.1	24.0	28.6	26.4	15	21.9	27.9	24.5	11
Body width	15.4	14.3	15.9	14.9	15	13.8	16.7	14.9	11
Head length	28.6	24.9	29.3	27.6	15	24.5	29.2	27.1	11
Snout length	8.0	6.5	8.3	7.5	15	6.7	9.1	7.8	11
Eye diameter	10.0	9.0	11.0	10.0	15	9.0	10.9	10.0	11
Interorbital width	11.9	11.2	12.8	12.1	15	11.2	12.9	12.1	11
Caudal-peduncle depth	11.5	11.4	12.7	12.0	15	10.3	12.8	11.6	11
Caudal-peduncle length	24.2	21.9	24.6	23.3	15	20.2	24.4	22.3	11
Pre-first dorsal distance	51.1	47.9	53.7	51.2	15	50.3	55.7	53.1	11
Pre-second dorsal distance	70.5	66.7	71.3	69.2	15	66.2	74.7	70.6	11
Preanal distance	64.8	58.8	67.5	62.4	15	58.8	65.4	61.6	11
Prepelvic distance	46.7	41.2	49.5	45.6	15	43.1	49.0	46.4	11
Base of second dorsal fin	13.2	11.8	14.9	13.3	15	9.3	15.3	11.5	11
Base of anal fin	18.2	16.7	19.6	18.3	15	16.0	19.7	17.8	11
Longest ray of 1st dorsal fin	24.7	15.7	28.9	21.5	15	10.6	14.9	13.3	11
Longest ray of 2nd dorsal fin	28.6	18.3	30.7	25.1	15	12.2	16.3	14.4	11
Longest ray of anal fin	31.2	18.6	36.1	26.8	15	12.8	19.0	15.5	11
Pectoral-fin length	23.0	19.1	23.1	21.8	15	17.9	22.8	20.2	11
Pelvic-fin length	16.8	14.1	19.3	16.8	15	11.9	15.6	13.7	11
Caudal-fin length	29.9	26.7	31.0	29.2	15	24.1	30.0	26.4	11

September 2014 (PU14-102); NTM S.17876-001, 9 specimens, 18.9–23.5 mm SL, Magazine Creek, Bowen, Queensland, 20° 00.03´S, 148° 15.08´E, P. Unmack, 16 September 2014, (PU14-108); NTM S.17877-012, 200 specimens, 9.0–25.5 mm SL, Sand Hills Creek, Bowen, Queensland, 19° 59.31´S, 148° 15.39´E, P. Unmack, 16 September 2014 (PU14-109); NTM S.18130-016, 14 specimens, 18.5–30.1 mm SL, collected with holotype; NTM S.18097-001, 7 specimens, 24.1–30.2 mm SL, saltwater creek at Bowen, Queensland, 19° 59.40´S, 148° 14.40´E, M. Lachlan, 2016; WAM P.35703-001, 12 specimens, 18.8–32.3 mm SL, collected with holotype.

Diagnosis: dorsal-fin elements IV–VI + 7–9 (usually V + 7 or 8, mean 7.8); anal-fin elements 9–12, mean 11.2; male second dorsal and anal fins moderately tall with middle rays longest and rounded profile, adpressed second dorsal fin of adult males usually overlapping caudal-fin base; longest ray of first dorsal fin of male 15.7–28.9, mean 21.5% SL; longest ray of second dorsal fin of male 18.3–30.7, mean 25.1% SL; longest anal ray of male 18.6–36.1, mean 26.8% SL; lateral scales 26–28 (rarely 26); transverse scales 6; predorsal scales 10–13 (usually 11–12), including 1–2 enlarged scales anteriorly; greatest body depth 21.9–28.6, mean 25.6% of SL; colour in life mainly semitransparent yellowish to pale greyish with dark scale outlines and series of pearly hash marks on row of scales immediately above mid-lateral row, mainly blackish dorsal fins, and bold black submarginal bands on upper and lower edges of caudal fin. Differs from the morphologically similar *P. signifer* genetically (congruent at nuclear and mitochondrial markers), and morphologically in having a blackish with pale margin dorsal fins vs black margin and smaller maximum size (about 33 mm vs 48 mm SL).

Description: dorsal-fin elements V+7 (IV–VI + 7–9, mean 5.1); anal-fin elements 11 (9–12, 11.2); pectoral-fin rays 12 (10–12, 11.2); pelvic-fin elements I,5; branched caudal rays 13 (13–14, 13.8); principal caudal rays 15 (15–16,15.8); accessory caudal rays 6 + 6 (5–6 + 5–7, 6 + 6); lateral scales 27 (26-28, 27.2); transverse scales 6; scales between dorsal fins 3 (1-3; 2.5); predorsal scales 11 (10–13, 11.7); preopercle scales 2 (2–3, 2.1); opercular scales 5 (4–6, 4.6); gill rakers on first arch 11 (10-14, 12.2).

Body depth 1.1 (1.0–1.2, 1.1); head length 3.5 (3.4–4.1, 3.7), pre-first-dorsal distance 2.0 (1.8–2.1, 1.9), pre-second-dorsal distance 1.4 (1.3–1.5, 1.4), preanal distance 1.5 (1.5–1.7, 1.6), prepelvic

Table 6. Frequency distribution of counts for fin rays, lateral and predorsal scales and gill rakers of *Pseudomugil halophilus*.

First dorsal spines				Second dorsal rays			Anal rays			
IV	**V**	**VI**	**VII**	**7**	**8**	**9**	**9**	**10**	**11**	**12**
2	20	3	1	9	14	3	1	3	12	10

Pectoral rays			Lateral scales			Predorsal scales			
10	**11**	**12**	**26**	**27**	**28**	**10**	**11**	**12**	**13**
2	16	8	3	15	8	1	8	15	2

Total gill rakers			
11	**12**	**13**	**14**
4	15	6	1

Figure 3. *Pseudomugil halophilus*, aquarium photographs, Bowen, Queensland, showing (upper image) small male (approximately 23 mm SL) and (lower image) similar-sized female in background. Photo: Michael Hammer

distance 2.1 (2.0–2.4, 2.2), all in SL. Greatest width of body 1.8 (1.5–1.9, 1.7) in body depth. Snout length 3.6 (3.0–4.2, 3.6); eye diameter 2.8 (2.4–3.0, 2.7); interorbital width 2.4 (2.0–2.4, 2.3); depth of caudal peduncle 2.5 (2.1–2.7, 2.3); length of caudal peduncle 1.2 (1.1–1.4, 1.2), all in HL.

First dorsal fin originates behind pelvic fin insertion by distance about equal to eye diameter; longest ray of first dorsal fin 1.2 (1.0–2.6, 1.6); second dorsal fin originates above middle of anal-fin base; longest ray of second dorsal fin 1.0 (0.9–2.3, 1.5), of anal fin 0.9 (0.7–2.0, 1.4), all in HL; all rays, except first, of second dorsal and anal fins branched, the last ray divided at base. Pelvic-fin length 1.7 (1.4–2.3, 1.8) in HL. Pectoral fins pointed, longest rays 1.7 (1.4–2.3, 1.8) in HL. Caudal fin forked, its length 3.3 (3.2–4.1, 3.6) in SL.

Jaws oblique, mouth terminal or lower jaw protruding when mouth open; posterior end of upper jaw not reaching to level of anterior edge of pupil. Anterior portion of upper jaw with patch of small villiform teeth with toothless gap between these and 3–4 enlarged, curved teeth projecting from lateral portion of posterior part of jaw. Lower jaw with several rows of small, curved villiform teeth anteriorly, narrowing to a single row posteriorly.

Scales cycloid, moderately large and elongated dorsoventrally, arranged in regular horizontal rows; predorsal scales extending to or almost to interorbital, terminating with 1–2 enlarged scales on frontal-interorbital; usually 2 (rarely 3) scales on cheek just below eye; cephalic sensory pores large and conspicuous, consisting of 5 interorbital pores, 6 postorbital/preopercular pores, and pair of preorbital pores on each side; anterior naris round with slightly elevated rim, situated near snout tip and posterior naris forming narrow slit adjacent to upper anterior edge of eye.

Colour in life and when fresh (Figs 3–4, see also photos by Wim Heemskerk in separate article appearing in this issue)**:** male overall semi-translucent yellowish to pale greyish with outline of swim bladder and stomach faintly visible; head pale yellowish except opercle with silvery reflections and pinkish hue imparted by underlying gills; iris silvery blue; conspicuous series of about 16 pearly hash marks on row of scales immediately above mid-lateral row, extending from just above pectoral-fin base to caudal-fin base, gradually decreasing in height and spaced at regular intervals, except last 4 larger, equal in size and more closely spaced; narrow, yellow mid-lateral stripe immediately below, nearly inconspicuous except on caudal peduncle, where equal in width to pearly spots just above; dorsal fins semi-translucent whitish on basal third to fourth and mainly blackish on remainder of fins with yellow tinge on posterior lobe of second dorsal fin; anal fin whitish basally and mainly semi-translucent yellowish on remainder of fin except a few inconspicuous blackish streaks on inter-radial membranes, usually on posterior half of fin; caudal fin mainly bright yellow medially with broad, blackish submarginal band along upper and lower edges, each with narrow white margin; pelvic fins semi-translucent whitish; pectoral fins translucent with thin black dorsal margin. Adult female generally plain compared to male with only slight darkening of the dorsal and anal fins and with less vivid submarginal dark bands on the caudal fin.

Colour of holotype in alcohol (Fig. 5)**:** overall yellowish white or tannish with pepper-like melanophores on scale margins (imparting overall reticulated appearance), more apparent on dorsal-most row, where expanded onto most of scale surface; predorsal scale row with strong concentration of melanophores, darkest centrally, forming blackish mid-dorsal stripe extending from first dorsal-fin origin to large postorbital scale; interorbital region and adjacent postorbital blackish due to concentration of melanophores; melanophores also concentrated on tip and side of snout (including upper lip) and on chin; cheek yellowish white; opercle yellowish white to silvery with clusters of faint-brown melanophores dorsally; double row of close-set melanophores forming midlateral blackish stripes extending from below level of first dorsal-fin origin to beginning of caudal peduncle, upper row darker and more conspicuous; dorsal and anal fins narrowly semi-translucent at base with membranous portion of most of fin blackish; caudal fin semi-translucent except about 5 upper and lowermost rays with blackish inter-radial membranes; pelvic fins semi-translucent.

Sexual dimorphism: based on comparisons of 13 males, 23.4–30.0 mm SL and 11 females, 22.1–32.3 mm SL. There are conspicuous differences between sexes in the length of the dorsal, anal, and pelvic fins as shown in Table 5. Males generally have longer fins than females, at least for individuals larger than about 23 mm SL. The adpressed first and second dorsal fins and pelvic fins of the male usually overlap or are at least level with the respective anterior bases of the second dorsal, caudal, and anal fins in contrast to the fins of the female that fall well short of these levels (Figs 3–4, 7).

Comparisons: the only species the new taxon is likely to be confused with is the similar appearing *P. signifer*, which ranges widely in coastal drainages of eastern Australia from Wonboyn Lake south of Eden, New South Wales (37° 15′S, 149° 56′E) to the islands of the Torres Strait, Queensland, extending southward on the western side of the Cape to the vicinity of Weipa (Saeed *et al.*, 1989). Although frequently occurring many km inland in pure fresh water (in the case of the Fitzroy River, as a disjunct population over 500 km from the river mouth), *P. signifer* also shares brackish mangrove habitats throughout the known range of *P. halophilus*. Regardless of the ultimate status of *P. signifer*, there is clear genetic separation from *P. halophilus* (Figs 1–2), which also differs in regard to certain male features, including colouration and shape of the dorsal and anal fins. There is also a notable difference in maximum size with *P. signifer*, occasionally reaching 55 mm SL, but usually about 40–46 mm SL compared with a maximum of about 33 mm SL for *P. halophilus*. These differences are summarised in Table 4.

Populations of *P. signifer* occurring north of Townsville, Queensland, frequently have exaggerated male dorsal and anal fins (Fig. 5) with long, trailing extensions involving the first dorsal-fin spine (to at least 56% SL), first segmented pelvic-fin ray (to at least 33% SL), and second ray of the second dorsal and anal fins (to at least 34 and 47% respectively). Their tall, triangular, sail-like dorsal and anal fins are immediately recognizable and not likely to be confused with those of *P. halophilus.* Although more southern populations that overlap the range of *P. halophilus* have shorter second dorsal and anal fins, these are generally angular in shape compared to the rounded fins of the new species. The shape differences reflect the relative lengths of the individual rays. In *P. signifer* the second ray of the second dorsal and anal fin is the longest with the remaining rays becoming progressively shorter. However, in *P. halophilus* the third to fifth rays of the second dorsal fin and fourth to sixth rays of the anal fin are longest and approximately subequal. Colouration of the male dorsal fins is also reliable for separating the two species, particularly when alive or freshly collected and viewed on a light-coloured background. The second dorsal fin of *P. halophilus* possesses considerable blackish pigment on the central portion of the fin compared to *P. signifer*, which has an intense black spot on the basal part of the anterior-most rays and blackish posterior margin (often present but much fainter in larger female *P. signifer* also). Moreover, the first dorsal fin of *P. halophilus* and posterior portion of the anal fin are also darkly pigmented compared to the fins of *P. signifer*, which are mainly pale except for the dark posterior margin of the anal fin.

Distribution and habitat: this species inhabits the lower reaches of coastal creeks of eastern Queensland, being recorded northward from the Gladstone area to the vicinity of Townsville (Fig. 1 and Table 1); however, it is likely they occur further south and north of this region. The habitat consists of small tidal creeks and drains within a few km of the sea, usually closely associated with mangroves. They were typically found along the margins of deeper habitats or in extensive shallow and more open sites (Figure 8). They are likely also found around the margins of larger habitats too like channels, but these remain poorly explored due to the likely presence of Saltwater Crocodiles (*Crocodylus porosus*).

Etymology: the species is named *halophilus* (Greek: “salt-loving”) with reference to its predilection for salt or brackish-water creeks.

Discussion

The Mangrove Blue-eye is a significant addition to the known Sahul freshwater fish fauna, bringing the total number of valid species in the family to 23 (including new *Kiunga* species, this issue). However, additional taxonomic work in several ‘species’ groups is ongoing (i.e. cryptic species complexes), and with

Figure 4. *Pseudomugil halophilus*, freshly collected paratypes, NTM S.18097-001, Bowen, Queensland: upper, male, 24.9 mm SL, middle, male, 26.4 mm SL, and lower, female 28.6 mm SL. Photo: Michael Hammer

new discoveries, the eventual number of species is likely to keep increasing. The current discovery occurred in a relatively well-studied region ichthyologically, and despite reviewing extensive museum material from previous sampling in search of the new species, only two previous records were uncovered. This is likely due to morphological similarity of *P. halophilus* to *P. signifer* and requiring a population level comparison (i.e. mature males). The very specific habitat of *P. halophilus* is also generally less targeted due to dense vegetative growth, thick mud, and the presence of Saltwater Crocodiles and biting insects. The initial key observations by a citizen scientists speak to the importance of aquarists in providing field and aquarium data to help inform understanding of biodiversity (Callaghan et al. 2021). While occurring over a relatively large range (~750 km), more detailed mapping to identifying key habitats and environmental requirements for protection of *P. halophilus* is required.

Figure 5. *Pseudomugil halophilus*, preserved holotype, 27.5 mm SL, QM I.41389, Gladstone, Queensland. Photo: Gerald Allen

The presence of two mitochondrial lineages within *P. signifer* (McGlashan & Hughes 2002; Wong et al. 2004) is curious given there is opportunity for unimpeded coastal gene flow between estuarine populations currently, or at least under lower sea level in the last glacial maxima with greater drainage connectivity. However, a high level of morphological variability in *P. signifer* was noted by Hadfield et al. (1979) and Saeed et al. (1989), who concluded that only a single species was represented; this result is also supported by preliminary findings using nuclear genetic markers in this study. Nevertheless, detailed genetic population sampling across the range using next generation sequencing is warranted. In the interim the use of the name *P. signatus* is discouraged, being retained here within synonymy of *P. signifer.*

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Figure 6. *Pseudomugil signifer*, aquarium photo, male, approximately 30 mm SL, near Innisfail, Queensland (G. Allen), and freshly collected male, 41 mm SL, NTM S.17957-002, Daintree River system, Queensland. Photo: Michael Hammer

Figure 7. Comparison of *Pseudomugil halophilus* (A&B: NTM. S.18130-016, paratypes) and *P. signifer* (C&D: NTM S.18130-015) collected together at Gladstone, Queensland type locality for *P. halophilus*: A. *P. halophilus*, male, 29.0 mm SL; B. *P. halophilus*, female, 25.7 mm SL; C. *P. signifer,* male, 32.4 mm SL; D. *P. signifer*, female, 29.4 mm SL Photo: Michael Hammer

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Figure 8. Habitat photos of *Pseudomugil halophilus* from Sand Hills Creek, Bowen (PU14-109), and Auckland Creek, Gladstone (PU15-17). Photos: Peter Unmack